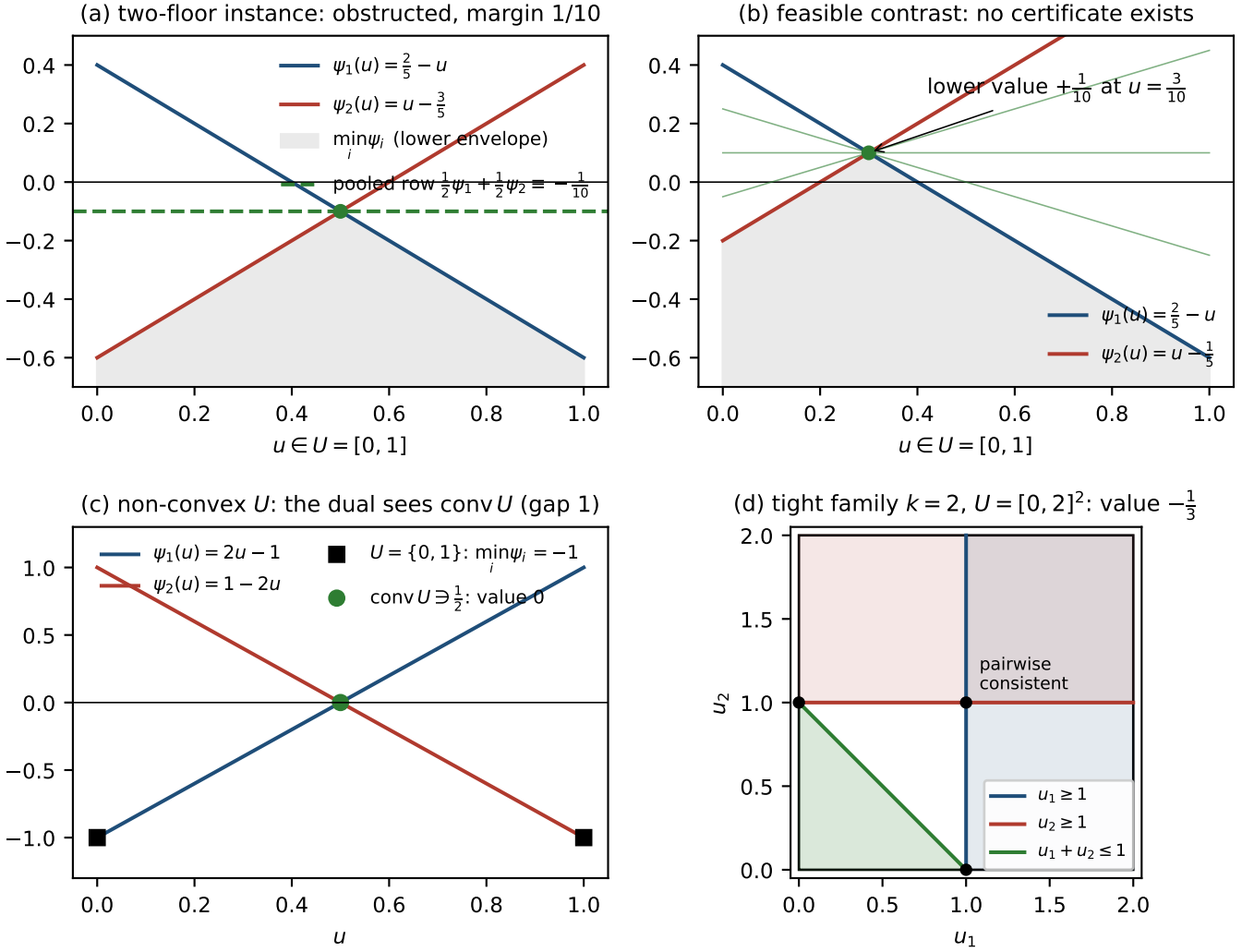
(a) two-floor instance: obstructed, margin 1/10
$\psi_1(u) = \frac{2}{5} - u$
$\psi_2(u) = u - \frac{3}{5}$
$\min_i \psi_i$ (lower envelope)
pooled row $\frac{1}{2}\psi_1 + \frac{1}{2}\psi_2 \equiv -\frac{1}{10}$
$u \in U = [0, 1]$
(b) feasible contrast: no certificate exists
lower value $+\frac{1}{10}$ at $u = \frac{3}{10}$
$\psi_1(u) = \frac{2}{5} - u$
$\psi_2(u) = u - \frac{1}{5}$
$u \in U = [0, 1]$
(c) non-convex U: the dual sees conv U (gap 1)
$\psi_1(u) = 2u - 1$
$\psi_2(u) = 1 - 2u$
$U = \{0, 1\}$: $\min_i \psi_i = -1$
conv $U \ni \frac{1}{2}$: value 0
u
(d) tight family $k = 2$, $U = [0, 2]^2$: value $-\frac{1}{3}$
pairwise consistent
$u_1 \geq 1$
$u_2 \geq 1$
$u_1 + u_2 \leq 1$
u_1
u_2